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SalUn-Guided Feature Visualization: Visualizing the Internal Changes of Neural Networks During Unlearning

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To all the people that led me here

Abstract

Machine unlearning—the selective removal of specific training data influence from neural networks—has emerged as a critical capability for privacy-compliant and trustworthy AI systems. This thesis presents a systematic investigation that identifies and resolves fundamental barriers to effective machine unlearning.

Through three experimental phases using InceptionV3 on ImageNet and ResNet-50 on TinyImageNet, we first establish baseline limitations of class-wise unlearning across individual domains, then validate these constraints across multiple semantic domains (cats, dogs, vehicles), and finally work with an alternative approach. Our comprehensive analysis reveals that class-wise unlearning consistently fails across different methods (Conservative, Gradient Ascent, Fine-tuning, Random Labeling) and domains, never exceeding 50% unlearning accuracy while often degrading model utility below 75%.

The key contribution of this work is the investigation of whether feature visualization can produce reliable proof of successful knowledge removal in machine unlearning. We also demonstrate that individual training samples have more localized neural representations that can be selectively modified, while semantic classes require changes that conflict with distributed knowledge organization.

Using SalUn (Saliency Unlearning) for systematic parameter modification, our approach achieves over 80% unlearning accuracy while preserving >98% model utility across all tested scenarios—representing a 60+ percentage point improvement over class-wise methods. Comprehensive weight analysis reveals that successful unlearning targets specific network components (batch normalization parameters, downsampling layers) rather than making arbitrary modifications, while feature visualization confirms selective knowledge removal without network degradation.

These findings establish machine unlearning as a practically viable technology for privacy compliance, data quality control, and model maintenance applications.

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Chapter 1

Introduction

1.1 Privacy imperatives in Deep Learning

Deep learning plays a critical role in enhancing privacy, integrity and security within the digital world. This has become a primary necessity in response to data protection regulations like "the right to be forgotten" , established by **The General Data Protection Regulation (GDPR)** [1], and others as a fundamental principle, mandating that companies remove personal data upon user request.

However, in the context of machine learning systems, complying with these standards raises major technical difficulties. When neural networks are trained on large datasets containing personal information, the learned parameters encode implicit knowledge about individual data points. Traditional approaches require complete model retraining from scratch, which is computationally very expensive in terms of resources and money.

In April 2021, the European Commission proposed the first **EU artificial intelligence law**, establishing a risk-based AI classification system. AI systems that can be used in different applications are analyzed and classified according to the risk they pose to users. The AI act, signed by the EU lawmakers in June 2024, sets a common framework for the use and supply of AI systems in the EU. It offers a classification for AI systems with different requirements and obligations. The regulation further establishes dedicated provisions for general-purpose AI (GPAI) systems, imposing tougher obligations on those GPAI models that possess “high-impact capabilities” , and could cause a systemic risk and have a significant impact on the internal market [2].

In addition to the AI act, we also have the **California Consumer Privacy Act CCPA** of 2018 [3], that gives consumers more control over the personal information that businesses collect about them. Even though it doesn’t directly address AI, it still highlights the importance of protecting the user’s data and their rights to limit the use and disclosure of sensitive information.

It is important to understand that Artificial Intelligence poses a range of real-world concerns that make regulations both necessary and challenging. Regulations like the **OECD AI Principles** are the first intergovernmental standard on AI: these principles offer a foundation for international cooperation and interoperability, in order to maximize the benefits of AI and minimize its risks [4].

1.2 Selective Knowledge Removal in Neural Networks

Machine unlearning (MU) emerges as a promising paradigm for bridging regulatory requirements and technical feasibility. It aims to remove the influence of a specific subset of training data points from trained models, such as private, wrong, manipulated data or even harmful content and misinformation, without needing to rebuild models from the ground up [5].

Generally speaking, the current machine unlearning methods can be divided into two categories: **exact** or **certified** MU and **approximate** MU. The former category has the goal of removing the influence of specific data points as if they were never part of the training process, and this can be achieved by retraining the model from scratch after removing the specific data, which is, as we said before, very resource-intensive.

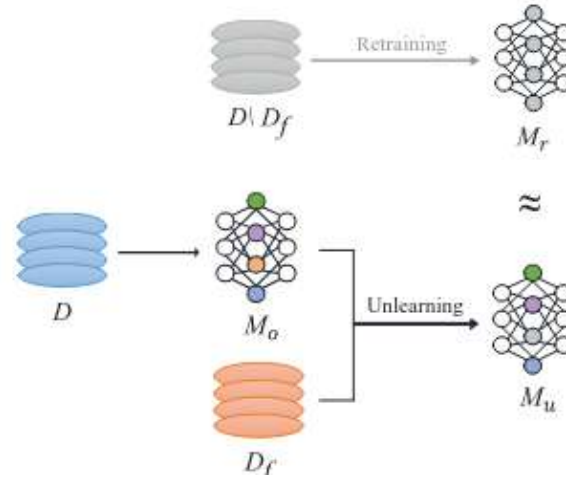


Figure 1.1. The objective of Machine unlearning. Adapted from [6]

The latter instead is a more practical approach, with the goal of reducing *the influence* of specific data points from the model, but with a trade-off in precision. While some machine unlearning techniques have shown potential in efficiently erasing data without full retraining, the challenge lies in maintaining model accuracy and performance at scale. As a matter of fact, a current focus of MU research is the trade-off between privacy and model utility, because ensuring that unlearning is complete and accurate is a complex task, requiring robust verification mechanisms. Another key aspect to consider is that, unlearning in large and complex models can be resource-intensive, so finding efficient ways to implement these complex algorithms is becoming increasingly important for energy conservation and greener practices.

There are various techniques to minimize the influence of unlearned data, such as *influence-based unlearning approaches* [7], which leverage influence function analysis of training data points [8], demonstrating high-performance variance and thereby

raising concerns about *instability* in approximate unlearning. Other methods also face similar limitations, such as **Fisher forgetting** [9] and **gradient ascent** [10].

These challenges become particularly pronounced in generative models for complex modalities, where toxic content can emerge from both explicit prompts and implicit combinations of safe concepts, necessitating novel approaches like Latent Code Replacement that operate training-free within discrete latent spaces [11].

Among approximate unlearning methods, **SalUn (Saliency Unlearning)** is a particularly promising approach that solves some of the stability concerns discussed previously [12]. SalUn uses gradient-based weight saliency analysis to determine the most relevant parameters for distinct data points or concept, allowing for focused parameter changes that reduce collateral damage to unrelated information. Differently from influence-based approaches, which are sensitive to hyperparameters, SalUn takes a more direct approach, producing saliency maps to emphasize which network weights contribute most crucially on data to be forgotten. SalUn has shown effective in both image classification and generation tasks, demonstrating its adaptability across domains.

1.3 Feature Visualization and Interpretability

There is a growing necessity for neural networks to be interpretable to humans, for a variety of reasons. To respond to these concerns, two major threads of research have begun to merge: **feature visualization** and **attribution** [13]. Feature visualization methods involve solving an optimization problem to find an input image that maximizes the activation of a target element, that can be either a neuron, a layer or a channel, in order to provide insights into what patterns neural networks tend to recognize and learn [14].



Figure 1.2. Feature visualization. Adapted from [13]

In the context of CNN's, feature visualization techniques range from filter/kernel visualization to CNN-Specific attribution methods, like Grad-CAM [15]. These

methods are specifically designed to leverage the spatial structure and hierarchical feature learning of CNN architectures.

The role of feature visualization becomes particularly important in the context of machine unlearning, bringing the fact that true unlearning should manifest visually in the learned representations, not just in the final predictions. In this thesis, the pre/post-unlearning comparison exhibits how the same input produces different activation patterns, providing direct evidence of parametric knowledge modification at the feature level.

The primary aim of this work is to investigate whether feature visualization can provide reliable intrinsic evidence of successful knowledge removal in machine unlearning.

Chapter 2

Related Works

The following chapter will first present the knowledge necessary to understand the aim of this work, then discuss the foundational research across the domains of this thesis, Machine Unlearning and Feature Visualization, establishing the current state of knowledge in each domain and identifying the gaps that motivate the integration of SalUn with a feature visualization analysis.

2.1 Image Classification on Convolutional Neural Networks

2.1.1 General Overview of CNNs

Convolutional Neural Networks (CNNs) are a specific type of neural networks that are primarily used in the field of pattern recognitions within images. This allows the encoding of image-specific features into the architecture, making the network more suited for image-focused tasks [16].

The layers of a CNN are:

- **Convolutional Layer:** acts as filters for images. finding out features from images.
- **Pooling Layer:** the extracted feature sets are then passed to ‘pooling layer’. This layer takes large images and shrink them down while preserving the most important information in them. It keeps the maximum value from each window, it preserves the best fits of each feature within the window.
- **Rectified Linear Unit Layer (ReLU):** it swaps every negative number of the pooling layer with 0. This helps the CNN stay mathematically stable by keeping learned values from getting stuck near 0 or blowing up toward infinity.
- **Fully Connected Layer:** the final layer is the fully connected layers which takes the high-level filtered images and translate them into categories with labels [17].

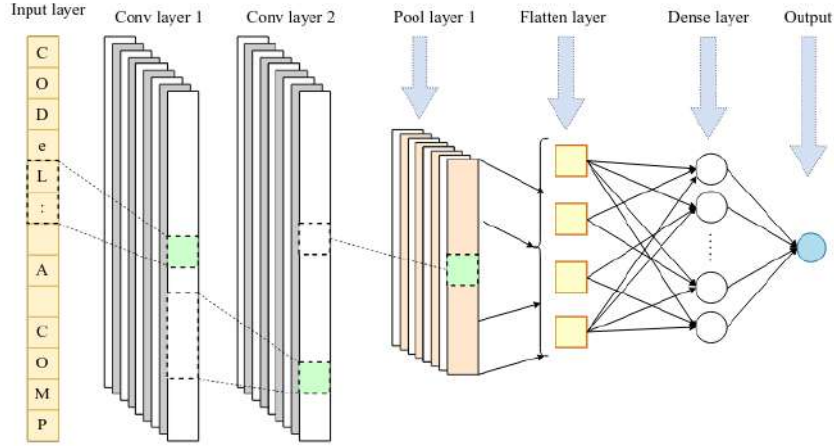


Figure 2.1. Simple one-dimensional CNN with two convolutional layers. From [18]

2.1.2 Image Classification on a CNN

Image classification is a supervised learning task where the goal is to assign a categorical label $y \in \{1, 2, \dots, C\}$ to an input image $X \in \mathbb{R}^{H \times W \times D}$, where C is the number of classes, H and W are spatial dimensions, and D represents channels (typically 3 for RGB images).

The CNN learns a mapping function $f : \mathbb{R}^{H \times W \times D} \rightarrow \mathbb{R}^C$ that outputs a probability distribution over classes:

$$P(y = c|X; \theta) = \frac{e^{f_c(X; \theta)}}{\sum_{j=1}^C e^{f_j(X; \theta)}} \quad (2.1)$$

where θ represents all learnable parameters in the network, and the softmax function ensures the outputs sum to 1.

CNNs excel at image classification due to their ability to learn hierarchical presentations automatically. The filters in the first layer learn to detect basic structural elements like edges and color blobs, while deeper layers are able to detect more complex compositional features [19].

2.2 Machine Unlearning

Machine unlearning approaches have become more popular in recent years, especially for deep neural networks where it is computationally costly to retrain from scratch. Because it allows the elimination of the effect of individual data points, entire data classes, or even higher-level data concepts from a pre-trained model without requiring complete retraining from scratch, it has emerged as a crucial paradigm in machine learning.

2.2.1 Problem Formulation

Let the training dataset be $D = \{z_i\}_{i=1}^N$, where each data point $z_i = (x_i, y_i)$ consists of features x_i and labels y_i in supervised learning. The subset of data to be removed is called the *forgetting dataset*, denoted $D_f \subseteq D$. Its complement, $D_r = D \setminus D_f$, is referred to as the *remaining dataset*.

Suppose the original model, trained on the full dataset D , is parameterized by θ_o . The retraining approach, denoted as *Retrain*, serves as the gold standard for machine unlearning: it involves training model parameters θ from scratch on the remaining dataset D_r . However, retraining is often computationally expensive and impractical for large-scale models.

The central challenge in machine unlearning is to derive an *unlearned model* with parameters θ_u from the original model θ_o and the forgetting dataset D_f (and/or the remaining dataset D_r), such that θ_u closely approximates the performance of Retrain, but is obtained in a significantly more efficient manner.

Formally, the goal is to construct θ_u such that:

$$\theta_u \approx \theta^* = \arg \min_{\theta} \frac{1}{|D_r|} \sum_{(x,y) \in D_r} \ell(f(x; \theta), y) \quad (2.2)$$

where ℓ is the loss function (e.g., cross-entropy), and $f(x; \theta)$ is the model prediction.

The effectiveness of machine unlearning is measured by how closely the unlearned model θ_u matches the performance and behavior of the retrained model θ^* on D_r , while ensuring that the influence of D_f is removed.

2.2.2 Machine Unlearning methods

Classical Baselines [20]

Fine-Tuning (FT). Fine-tuning continues training the original model θ_o exclusively on the retain set D_r for a specified number of epochs, which is treated as a hyper-parameter and may vary depending on the dataset and model architecture. The training objective is to minimize the cross-entropy loss over the retain set:

$$\mathcal{L}_{\text{FT}} = -\frac{1}{|D_r|} \sum_{(x,y) \in D_r} \log p(y|x; \theta) \quad (2.3)$$

Successive Random Labels (SRL). In the SRL approach, the model θ_o is trained on both the forget set D_f and the retain set D_r , but the labels of D_f are randomly reassigned at each epoch. Specifically, for each epoch, the labels for D_f are drawn independently from a uniform distribution $\mathcal{U}_C$ over the C possible classes. The training objective is:

$$\mathcal{L}_{\text{SRL}} = -\frac{1}{|D_r| + |D_f|} \left(\sum_{(x,y) \in D_r} \log p(y|x; \theta) + \sum_{(x,\tilde{y}) \in D_f} \log p(\tilde{y}|x; \theta) \right) \quad (2.4)$$

where $\tilde{y} \sim \mathcal{U}_C$.

Gradient Ascent (GA). Gradient ascent unlearning updates the model by performing gradient ascent steps on the forget set D_F . For each batch from D_F , the cross-entropy loss is computed and the model parameters are updated in the direction that increases this loss, rather than minimizing it. The update rule for parameters θ is:

$$\theta \leftarrow \theta + \eta \nabla_{\theta} \mathcal{L}_{\text{CE}}(D_F; \theta) \quad (2.5)$$

where η is the learning rate and $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss over D_F .

State-of-the-art MU methods

Fisher Forgetting (FF) Fisher Forgetting uses the Fisher Information Matrix (FIM) to estimate parameter importance. The diagonal of the FIM for parameter θ_i is:

$$F_{ii} = \mathbb{E}_{(x,y) \sim D} \left[\left(\frac{\partial \log p(y|x; \theta)}{\partial \theta_i} \right)^2 \right] \quad (2.6)$$

Parameters with high F_{ii} are considered important and are either masked or regularized during unlearning, while less important parameters are updated to forget D_f [9].

Influence Unlearning (IU) Influence-based methods estimate the effect of removing a data point z from the training set using influence functions:

$$\theta_{-z} \approx \theta - H^{-1} \nabla_{\theta} \ell(z, \theta) \quad (2.7)$$

where H is the Hessian of the loss over D , and ℓ is the loss for z . This approach is principled but computationally expensive for deep models [8].

Bad Teacher (BT) The Bad Teacher method employs a teacher-student framework to induce unlearning. The original model, referred to as the *good teacher* with parameters θ_{good} , is trained on the full dataset D . A *bad teacher* model, θ_{bad} , is initialized randomly and is intended to represent the knowledge to be forgotten.

The student model, θ_{student} , is initialized as a copy of the good teacher:

$$\theta_{\text{student}}^{(0)} = \theta_{\text{good}} \quad (2.8)$$

The student model is optimized to minimize $\mathcal{L}_{\text{BT}}$:

$$\theta_{\text{student}}^* = \arg \min_{\theta} \mathcal{L}_{\text{BT}} \quad (2.9)$$

This approach encourages the student to retain knowledge from the good teacher on the retain set while adopting the bad teacher’s behavior on the forget set, effectively unlearning the targeted data [21].

Saliency Unlearning (SalUn) SalUn leverages gradient-based weight saliency to identify and update the model parameters most responsible for encoding the forget set D_f . For each parameter θ_i , the saliency is computed as:

$$S_{\theta_i} = \left| \frac{\partial \mathcal{L}_{D_f}}{\partial \theta_i} \right| \quad (2.10)$$

where $\mathcal{L}_{D_f}$ is the loss over the forget set. Parameters with high S_{θ_i} are targeted for modification. The model is then updated by minimizing the loss on the retain set D_r and maximizing (or randomizing) the loss on D_f , often using random labels for D_f to ensure effective forgetting [12].

2.3 Interpretability

Interpretability is the study of explaining models in understandable terms to humans. Currently, interpretability is divided into two paradigms: **the intrinsic paradigm**, which believes that only models designed to be explained can be explained, and **the post-hoc paradigm**, which believes that black-box models cannot be explained [22].

The Intrinsic Paradigm These models are architecturally constrained, such that the explanation emerges from the architecture itself.

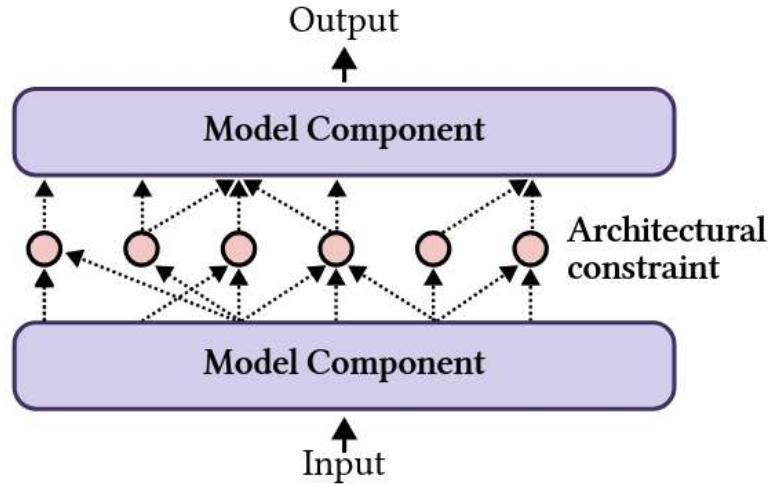


Figure 2.2. Abstract diagram of the intrinsic paradigm. From [22]

The Post-Hoc Paradigm The model is produced without regard for explanation, and the explanations are then created after model training. However, there are often some simple criteria, like “the model should be differentiable”, “the training dataset is known”, or “inputs are represented as tokens”. Although general applicability is technically not a requirement, if a method is so specific that it only works on one specific model, it’s likely an intrinsic explanation.

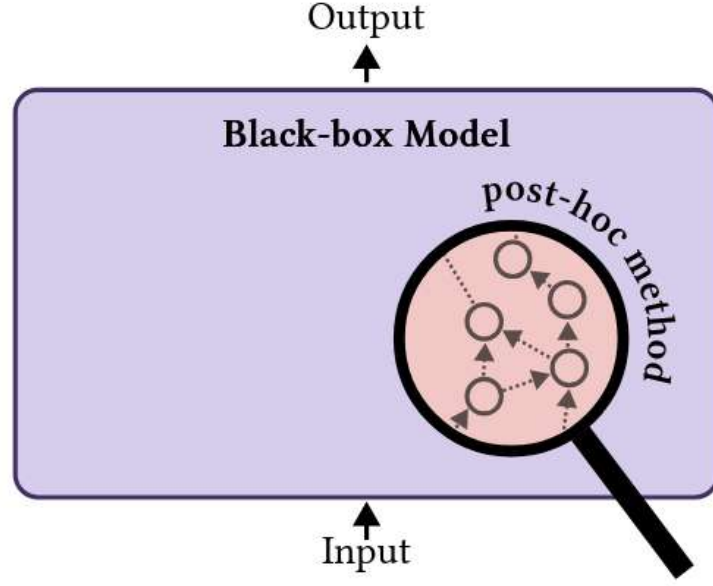


Figure 2.3. Abstract diagram of the post-hoc paradigm. From [22]

2.3.1 Feature Visualization

Feature visualization is a powerful post-hoc interpretability tool that reveals what internal units (neurons, channel, or layers in a neural network) respond, helping us understand the representations learned by the model. Unlike attribution methods (e.g., saliency maps, Grad-CAM) that show where a model focuses in a specific input, feature visualization synthesizes an input that maximally activates a target feature, providing insight into what patterns the network has learned [23].

Activation Maximization It seeks to synthesize an input that maximally activates a target neuron or feature. It is formulated as a non-convex optimization problem:

$$x^* = \arg \max_x (a(x) - R(x)) \quad (2.11)$$

where $a(x)$ is the activation of the target neuron for input x , and $R(x)$ is a regularization term (e.g., total variation, L_2 norm, or other priors) to encourage natural-looking images.

The optimization is typically performed via gradient ascent, with the update rule:

$$x_{t+1} = x_t + \epsilon_1 \frac{\partial a(x_t)}{\partial x_t} - \epsilon_2 \frac{\partial R(x_t)}{\partial x_t} \quad (2.12)$$

where ϵ_1 and ϵ_2 are step sizes for the activation and regularization terms, respectively.

2.3.2 Diversity in Feature Visualization

Two critical challenges in activation maximization are achieving **diversity** in generated visualizations and ensuring **selectivity** for the target feature. These

challenges arise from the multifaceted nature of neurons and the optimization properties of neural networks.

The Diversity Challenge Neurons in deep networks are often multifaceted, responding strongly to multiple distinct types of stimuli or *facets* [24]. For example, a face-detecting unit in AlexNet has been observed to respond to both human and lion faces. Moreover, activation maximization starting from different random initializations often converges to similar results.

Several techniques have been developed to improve visualization diversity:

- **Neural Path Dropout:** Randomly dropping certain neural paths during backpropagation to produce different facets of the same neuron [25].
- **Cluster-based Initialization:** Clustering training set images into groups and initializing optimization from average images computed from each cluster [24].
- **Distance Maximization:** Enforcing diversity by maximizing distance (e.g., cosine similarity) in pixel space between a reference image and the one being synthesized:

$$x^* = \arg \max_x (a(x) - R(x) - \lambda \cdot \text{sim}(x, x_{\text{ref}})) \quad (2.13)$$

where $\text{sim}(\cdot, \cdot)$ is a similarity metric and λ controls the diversity penalty.

- **Multi-neuron Activation:** Simultaneously activating pairs of neurons to reveal different facets. For instance, activating both "castle" and "candle" neurons produces castle-on-fire imagery, while "piano" and "candle" yields candle-on-piano visualizations [26].
- **Noise Injection:** Adding random noise at each optimization step to encourage exploration of different modes:

$$x_{t+1} = x_t + \epsilon_1 \frac{\partial a(x_t)}{\partial x_t} - \epsilon_2 \frac{\partial R(x_t)}{\partial x_t} + \mathcal{N}(0, \sigma^2) \quad (2.14)$$

The Selectivity Challenge Standard activation maximization may inadvertently activate other neurons in the same layer more strongly than the target neuron. This occurs because maximizing raw activation $a_i(x)$ for neuron i does not guarantee that neuron i will have the highest activation across the layer [27].

To enforce selectivity, researchers maximize the softmax or log-softmax of activations across a layer rather than raw activations. The selective objective becomes:

$$x^* = \arg \max_x \log \left(\frac{e^{a_i(x)}}{\sum_j e^{a_j(x)}} \right) - R(x) \quad (2.15)$$

where the summation is over all neurons j in the same layer as the target neuron i .

This approach offers several advantages:

- **Guaranteed Selectivity:** The target neuron is assured to be the most highly activated in its layer.
- **Interpretability:** Visualizations contain features exclusively relevant to the target neuron.
- **Theoretical Justification:** The log-softmax formulation aligns with probabilistic interpretations of activation maximization [27].

This framework for diversity and selectivity provides the foundation for robust feature visualization analysis in machine unlearning applications, ensuring that visual evidence of knowledge removal is both comprehensive and precise.

Chapter 3

Methodology

This chapter defines the methodological approach for investigating machine unlearning through feature visualization, presenting the experimental setup, models, datasets and evaluation metrics. This work presents the two main categories of MU for image classification: *class-wise forgetting* and *random data forgetting*. But first, let's explain in detail the main components of this thesis: **SalUn** and **Lucent**.

3.1 Unlearning via Gradient-based Weight Saliency

3.1.1 Gradient-based weight saliency map

The concept of gradient-based weight saliency mapping represents a fundamental change from traditional whole-model unlearning approaches toward targeted, parameter-specific modifications. It draws inspiration from the established principle of gradient-based input saliency maps [28, 29], which find the most influential input features for model predictions. Similarly, weight saliency mapping seeks to identify which model parameters are most critical for encoding knowledge about the data to be forgotten.

This modularity enables selective modification of parameters while preserving of knowledge unrelated to the forget set. Formally, the gradient-based weight saliency map is defined as:

$$\mathbf{m}_S = \mathbf{1} (|\nabla_{\theta} \ell_f(\theta; \mathcal{D}_f)|_{\theta=\theta_o}| \geq \gamma) \quad (3.1)$$

where:

- $\ell_f(\theta; \mathcal{D}_f)$ is the loss function defined on the forget set $\mathcal{D}_f$,
- $\nabla_{\theta} \ell_f$ denotes the gradient with respect to the model parameters θ evaluated at $\theta = \theta_o$,
- $|\cdot|$ is the element-wise absolute value,
- γ is a specified threshold (often the median of the gradient magnitudes),
- $\mathbf{1}(\cdot)$ is the indicator function that yields 1 when the condition holds and 0 otherwise,
- $\mathbf{m}_S$ is a binary mask highlighting the most salient weights for unlearning.

Based on (3.1), the unlearning model θ_u can be explicitly expressed as:

$$\theta_u = \mathbf{m}_S \odot (\Delta\theta + \theta_o) + (1 - \mathbf{m}_S) \odot \theta_o \quad (3.2)$$

where:

- θ_u are the updated (unlearned) model weights,
- $\mathbf{m}_S$ is the binary saliency mask identifying salient weights,
- $\Delta\theta$ denotes the weight changes computed during targeted unlearning,
- θ_o is the original pre-unlearning weight vector,
- $\odot$ is element-wise multiplication.

This selective mechanism ensures that unlearning is focused on crucial parameters, minimizing unwanted changes elsewhere in the model and preserving retained knowledge.

3.1.2 SalUn

SalUn incorporates (3.2) into the unlearning process. One of its advantages is its plug-and-play capability, allowing it to be applied on top of other MU methods. Actually, it has been proved that integrating weight saliency with RL (random labeling) shows very promising results [12].

In the context of image classification, Random Labeling (RL) involves assigning a random label to a data point intended for forgetting and subsequently fine-tuning the model using the randomly labeled forget set D_f . Within the SalUn framework, RL is employed to update the salient weights via the optimization problem given by:

$$\begin{aligned} \min_{\Delta\theta} L_{\text{SalUn}}^{(1)}(\theta_u) := & \mathbb{E}_{(x,y) \sim D_f, y' \neq y} [\ell_{\text{CE}}(\theta_u; \mathbf{x}, y')] \\ & + \alpha \mathbb{E}_{(x,y) \sim D_r} [\ell_{\text{CE}}(\theta_u; \mathbf{x}, y)] \end{aligned} \quad (3.3)$$

Here, y' represents a random label for the image x , distinct from its true label y , and θ_u is defined as previously in (3.2). The second term in the loss function enforces regularization on the retain set D_r with weight $\alpha > 0$, balancing the model's forgetting capability on the forget set with the retention of generalization performance on the non-forgetting data.

3.2 Feature Visualization with Lucent

To examine the impact of unlearning and the internal representations, we used Lucent, an open-source PyTorch library for feature visualization. It is inspired by the original Lucid library for TensorFlow.

Through **Activation Maximization**, discussed in the previous chapter, Lucent creates visuals that strongly stimulate a selected neuron, channel, or layer in a trained model. Its core principles come from the Distill article by Olah et al. [*Feature*

Visualization" (Distill, 2017)] [13], which describes how input images can reveal the "preferred stimuli" for units in deep neural networks.

Following this idea, Lucent:

- 1 Starts with a random image (noise)
- 2 Iteratively optimizes the input using GA to maximize a target activation
- 3 Regularizes the optimization, using transformations, to encourage interpretable images

In this thesis, Lucent was used to examine:

- **Before and after unlearning:** visualizing the same neurons or channels in the original and unlearned models, to assess if they were altered.
- **Layer-wise analysis:** feature visualizations at several networks depths, to trace how unlearning altered the abstraction in learned representations.

3.3 Experimental Design Overview

This work follows a sort of iterative experimental approach to investigate machine unlearning through feature visualization.

- 1 **Phase I:** class-wise forgetting MU (InceptionV3 + ImageNet)
- 2 **Phase II:** multi-domain class-wise forgetting (ResNet-50 + TinyImageNet)
- 3 **Phase III:** random data forgetting with comprehensive analysis (ResNet-50 + TinyImageNet)

We will quickly go through the first two phases and primarily focus on phase III.

3.3.1 Experimental Setup

All experiments were executed on a dual-socket server with the following specifications:

- **CPU:** Intel(R) Xeon(R) Gold 6136 (12 physical cores and 24 threads, 3.00 GHz)
- **Memory:** 256 GB of RAM
- **GPU:** NVIDIA Quadro P6000 (24 GB VRAM)

3.3.2 Dataset selection

- **ImageNet (Phase I):** For the first class-wise forgetting experiments, we utilized the entire ImageNet dataset. This choice was motivated by the rich semantic diversity and high-resolution images (224x224 pixels). The classes to forget were five cat-related classes (Egyptian cat, Tabby cat, Tiger cat, Persian cat and Siamese cat).
- **TinyImageNet (Phase II and III):** The transition to this dataset (64x64 pixels, 200 classes) was driven by computational efficiency considerations and the need for rapid experimental iteration.

3.3.3 Model Selection

- **InceptionV3:** It has approximately 23.8 million parameters, used in the beginning to get close to the Lucent notebooks' results (they used InceptionV1 but it is not available in Pytorch anymore).
- **ResNet-50:** It has approximately 25.6 million parameters, simpler architecture compared to InceptionV3, achieves higher accuracy and is more robust.

3.3.4 Phase I and Phase II

For these parts of the project, we modified the original SalUn implementation to support class-wise forgetting on image classification. This adaptation involved:

- **Loading the class mask:** it specifies which samples belong to class-to-forget and prepares the dataset labels for the unlearning algorithm.
- **Splitting data by class:** using the mask, the dataset's indices get partitioned into `forget_ids` (for the target class) and `retain_ids` (for all other classes).
- **Class-aware test set evaluation:** the validation/test set is split based on *class IDs*.

3.3.5 Phase III: random data forgetting and weight analysis

Objective: Conduct in-depth analysis using standard SalUn protocol with systematic weight analysis and visualization.

- Forgetting percentages: 10%, 20% and 30% of training data
- Layer-wise, channel-wise, and weight-specific impact analysis

Analyzing the impact of unlearning

This analysis is designed to compare the baseline (original) model against the unlearned model to understand what changed internally. It consists of:

- One script containing a class that provides three levels of granularity for comparing the unlearned model to the baseline model in a methodical and hierarchical way:

- 1 **Layer-level**: identifies the most affected layers overall by computing the mean relative change between weight tensors.
 - 2 **Channel-level**: isolates individual channels to see if changes were localized.
 - 3 **Weight-level**: Pinpoints the specific weight parameters within the top channels that underwent the most significant modification.
- Another script that extends the analysis to a comparative framework across multiple unlearning scenarios. It is designed to study experiments with the different forgetting percentages. It runs the class mentioned previously on each experiment and then performs a meta-analysis to identify trends such as correlating the forgetting ratio with the magnitude of parameter changes and finding layers that are commonly affected across different unlearning intensities.

3.3.6 Evaluation metrics

Performance-based metrics

- **Forget Accuracy (FA)**: The classification accuracy of the unlearned model on the forget set D_f . For effective unlearning, the metric should decrease significantly.
- **Retain Accuracy (RA)**: The classification accuracy on the retain set D_r , which should be preserved after unlearning to maintain model performance [20].
- **Unlearning Accuracy (UA)**: We define $UA(\theta_u) = 1 - Acc_{D_f}(\theta_u)$ to characterize the efficacy of machine unlearning in the accuracy dimension, where $Acc_{D_f}(\theta_u)$ is the accuracy of θ_u on the forgetting dataset D_f . It is important to note that a more favorable UA for an approximate unlearning method should reduce its performance disparity with the gold-standard retrained model (Retrain); a higher value is not necessarily better [30].
- **Testing accuracy (TA)**: This measure the *generalization ability* of θ_u on a testing dataset rather than D_f or D_r . TA is evaluated on the whole test dataset, except for class-wise forgetting, in which testing data points belonging to the forgetting class are not in the testing scope.

Quantifying structural unlearning impact We use two key metrics to quantify the impact of machine unlearning on the features of the model:

- **The Mean Absolute Change (MAC)**, defined as:

$$MAC(\mathbf{W}) = \frac{1}{N} \sum_{i=1}^N |\mathbf{W}_{\text{baseline},i} - \mathbf{W}_{\text{unlearned},i}|$$

- **The Mean Relative Change (MRC)**, which normalizes the change by the original parameter’s magnitude:

$$\text{MRC}(\mathbf{W}) = \frac{1}{N} \sum_{i=1}^N \frac{|\mathbf{W}_{\text{baseline},i} - \mathbf{W}_{\text{unlearned},i}|}{|\mathbf{W}_{\text{baseline},i}| + \epsilon}$$

Chapter 4

Results

This chapter presents the results of our three-phase experimental investigation into machine unlearning with feature visualization. We started with exploratory phases (phases I and II) that revealed important class-wise forgetting issues. Phase III (random data forgetting) was a thorough investigation that produced comprehensive results and new insights into the inner workings of machine unlearning.

4.1 Phase I: Initial class-wise forgetting with InceptionV3

Our experiments evaluated two saliency mask thresholds (0.3 and 0.5) to determine optimal weight selection strategies:

Mask	Retain $\uparrow$ (%)	Forget $\downarrow$ (%)	Unlearn $\uparrow$ (%)
0.3	63.38	53.17	46.83
0.5	77.51	66.74	33.26

Table 4.1. Phase I class-wise forgetting performance

Key Observations:

- **Threshold sensitivity:** Different mask thresholds produce notably different results
- **Trade-off relationship:** Lower threshold (0.3) achieves better unlearning accuracy (46.83%) but with reduced retain accuracy (63.38%)
- **Higher threshold (0.5):** Better preserves retain accuracy (77.51%) but achieves lower unlearning effectiveness (33.26%)
- **Forget accuracy patterns:** Both configurations show partial knowledge removal but incomplete forgetting

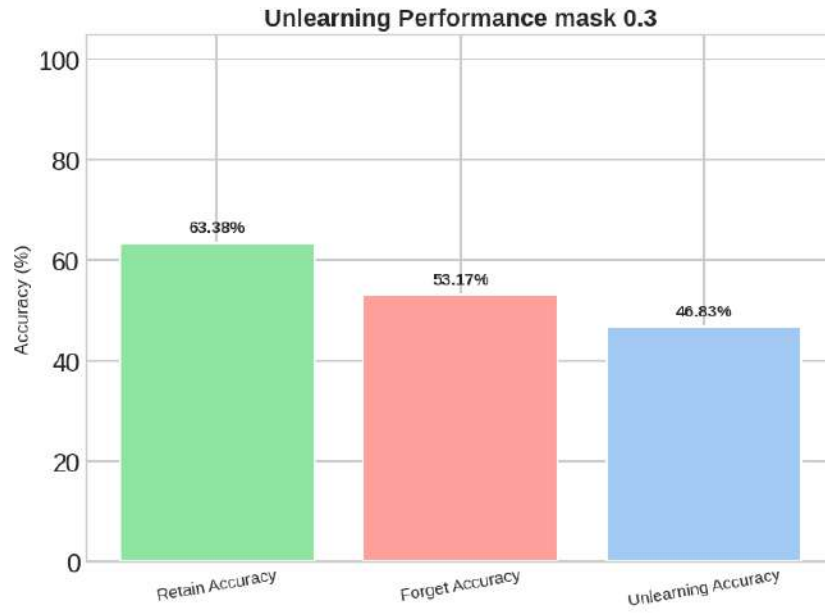


Figure 4.1. Forgetting cats with 0.3 mask

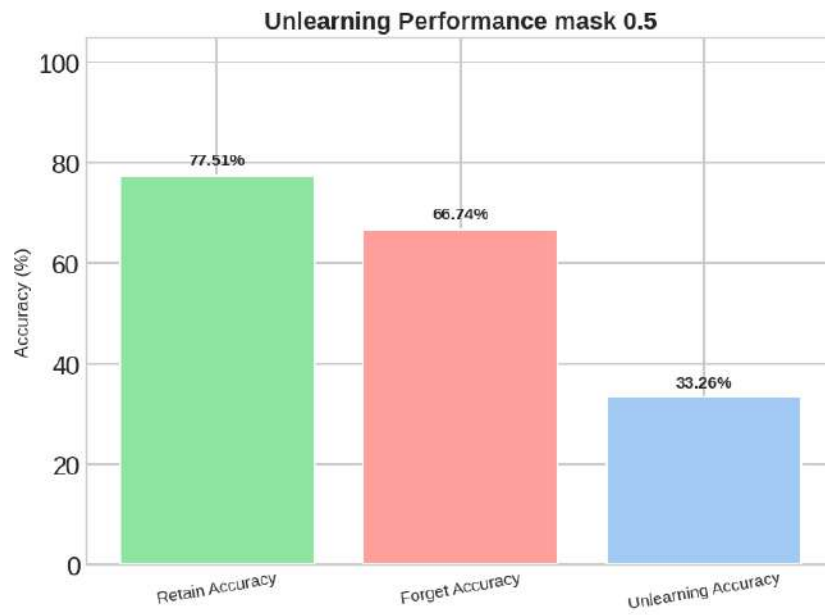


Figure 4.2. Forgetting cats with 0.5 mask

The results reveal a clear **trade-off relationship** between unlearning effectiveness and knowledge retention.

4.1.1 Mask Threshold 0.3 Analysis

- Selects more weights for modification (lower threshold = more salient weights)

- Achieves better unlearning accuracy (46.83% vs 33.26%)
- Causes more disruption to retain performance (63.38% vs 77.51%)
- Suggests broader weight modification impacts both target and non-target knowledge

4.1.2 Mask Threshold 0.5 Analysis

- More selective weight modification (higher threshold = fewer salient weights)
- Better preserve non-target knowledge (retain accuracy 77.51%).
- Less effective at removing target knowledge (unlearning accuracy 33.26%).
- Indicates that highly salient weights alone may be insufficient for complete forgetting

Saliency Threshold Sensitivity The significant performance differences across thresholds provide important insights:

1. **Weight Selection Granularity:** The threshold choice critically impacts which weights are modified
2. **Distributed Representations:** Class-specific knowledge appears distributed across weights with varying saliency levels
3. **Collateral Damage:** More aggressive weight selection (lower threshold) causes broader network disruption
4. **Incomplete Targeting:** Even selective approaches (higher threshold) fail to achieve complete forgetting

Class-wise Forgetting Challenges

Both configurations demonstrate inherent difficulties with semantic-level unlearning:

- Cat-specific features resist clean removal without affecting related concepts
- The gradient-based saliency may inadequately capture all relevant class-specific weights
- InceptionV3’s complex architecture distributes semantic knowledge across multiple pathways

4.1.3 Feature Visualization Analysis

To examine unlearning effects at different network depths, we conducted systematic visualization analysis across InceptionV3 layers. The mixed6e layer, representing high-level feature extraction before classification, provides particularly revealing insights into the unlearning process.



(a) Pre-unlearning: Rich, coherent facial and textural patterns (b) Post-unlearning: Fragmented patterns with reduced coherence

Figure 4.3. Feature visualizations from InceptionV3 mixed6e layer (channel 200) demonstrating unlearning impact on high-level semantic features

High-Level Feature Modification The mixed6e layer analysis reveals interesting changes in semantic feature representation:

Pre-unlearning characteristics (Figure 4.3a):

- Clear facial structure detection with detailed fur texture patterns
- Coherent animal-like features consistent with cat-class targeting
- Rich color gradients and well-defined boundaries
- High-level semantic integration of multiple visual elements

Post-unlearning modifications (Figure 4.3b):

- Fragmentation of coherent facial structures
- Loss of clear animal-specific features
- Reduced color organization and boundary definition
- Transformation from semantic to more abstract pattern recognition

Implications for Class-wise Forgetting The mixed6e layer changes provide crucial insights into the unlearning mechanism:

1. **Semantic Feature Disruption:** The transformation from recognizable animal features to abstract patterns indicates successful targeting of class-specific representations
2. **Hierarchical Impact:** High-level layers show more pronounced semantic changes compared to lower-level texture modifications
3. **Incomplete Removal:** Retention of some structural elements suggests partial rather than complete feature elimination
4. **Class Specificity:** The shift away from animal-like patterns aligns with cat-class forgetting objectives

Diversity and Unlearning We systematically analyzed the Mixed6b layer (channel 0) using Lucent’s feature visualization capabilities with diversity regularization to ensure robust feature extraction.

Visualization Parameters:

- **Target:** Mixed6b layer, channel 0
- **Objective:** Channel optimization with diversity regularization ($\lambda = 10^3$)
- **Diversity Weight:** 10^3 to encourage varied feature representations
- **Optimization:** Standard Lucent parameters with batch processing

Diversity Analysis Results

Figure 4.4 demonstrates the profound impact of class-wise unlearning on feature diversity within the Mixed6b layer. The systematic comparison across four diversity samples reveals distinct patterns of representational change.

Pre-unlearning Feature Characteristics The original model (top row) exhibits rich, coherent feature diversity:

- **Sample 1 (a):** Radially organized dot patterns with consistent spacing and natural color variation
- **Sample 2 (b):** Complex geometric grid structures with intersecting diagonal patterns
- **Sample 3 (c):** Organic cellular formations with vibrant green-purple coloration
- **Sample 4 (d):** Feather-like branching patterns with fine textural detail

These patterns demonstrate the layer’s capacity for diverse, semantically rich feature detection suitable for complex visual recognition tasks.

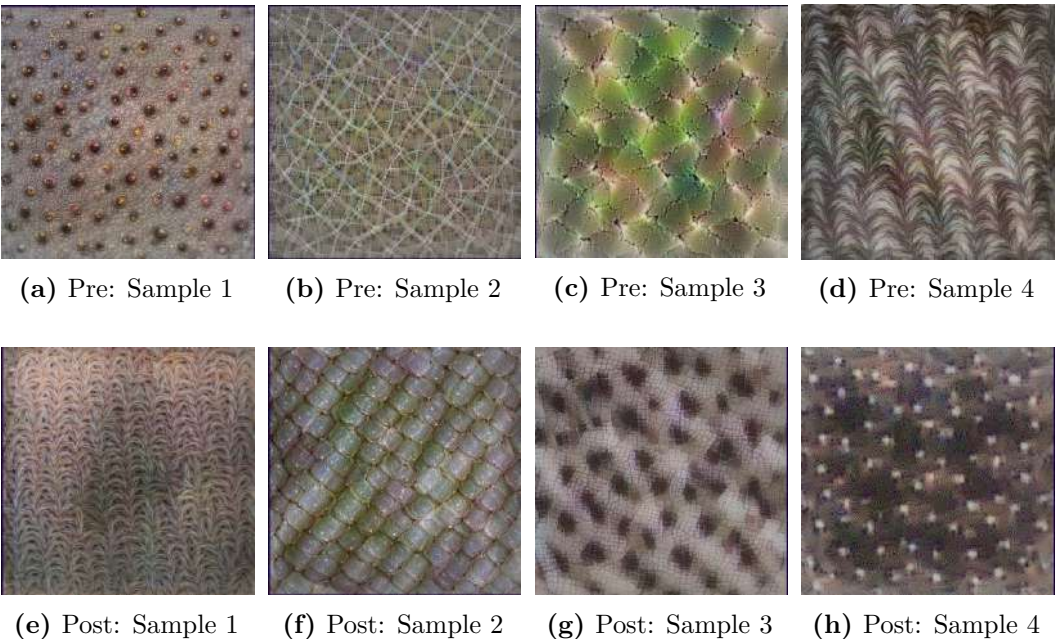


Figure 4.4. Mixed6b layer diversity analysis showing pre-unlearning (top row) and post-unlearning (bottom row) feature representations

Post-unlearning Degradation Patterns The unlearned model (bottom row) shows systematic feature degradation:

- **Sample 1 (e):** Loss of radial organization, emergence of wave-like distortions
- **Sample 2 (f):** Fragmented grid structure with reduced geometric coherence
- **Sample 3 (g):** Severe degradation into noise-like patterns with lost color organization
- **Sample 4 (h):** Simplified dot patterns with significantly reduced structural complexity

Sample	Pattern Coherence	Color Organization	Structural Integrity
1	Severely reduced	Maintained	Distorted
2	Fragmented	Partially lost	Compromised
3	Completely lost	Destroyed	Noise-like
4	Simplified	Reduced	Minimal structure

Table 4.2. Qualitative assessment of unlearning impact across diversity samples

Quantitative Impact Assessment

Implications for Class-wise Unlearning The diversity analysis reveals several critical insights:

1. **Non-uniform Impact:** Different diversity samples show varying degrees of degradation, indicating selective rather than uniform feature destruction
2. **Pattern-specific Effects:** Organic patterns (Sample 3) show more severe degradation than geometric ones (Sample 2), potentially reflecting the biological nature of target classes (cats)
3. **Structural vs. Textural Changes:** Some samples maintain overall structure while losing fine detail (Sample 1), while others undergo complete organizational collapse (Sample 3)
4. **Preserved Diversity Mechanism:** Despite degradation, the diversity regularization continues to produce distinct patterns, indicating that the unlearning process affects feature content rather than the diversity generation mechanism itself

Evidence of Targeted Knowledge Removal The systematic degradation across all diversity samples provides compelling evidence for successful knowledge modification:

- **Consistent Direction:** All samples show degradation rather than random changes, indicating purposeful modification
- **Preserved Functionality:** The layer continues to generate diverse patterns, suggesting selective rather than destructive unlearning
- **Semantic Relevance:** The most severe degradation occurs in organic, cellular patterns (Sample 3) that align with animal-specific features relevant to cat class forgetting

This comprehensive diversity analysis demonstrates that the unlearning process successfully modifies deep semantic features while maintaining the fundamental diversity-generation capabilities of the network layer.

4.2 Phase II: Multi-Domain Class Forgetting Analysis

Building on Phase I findings, we extended the class-wise forgetting investigation across multiple domains using ResNet-50 on TinyImageNet. This expansion aimed to determine whether the observed limitations were cat-specific or represented broader challenges in class-wise unlearning.

4.2.1 Cross-Domain Performance Analysis

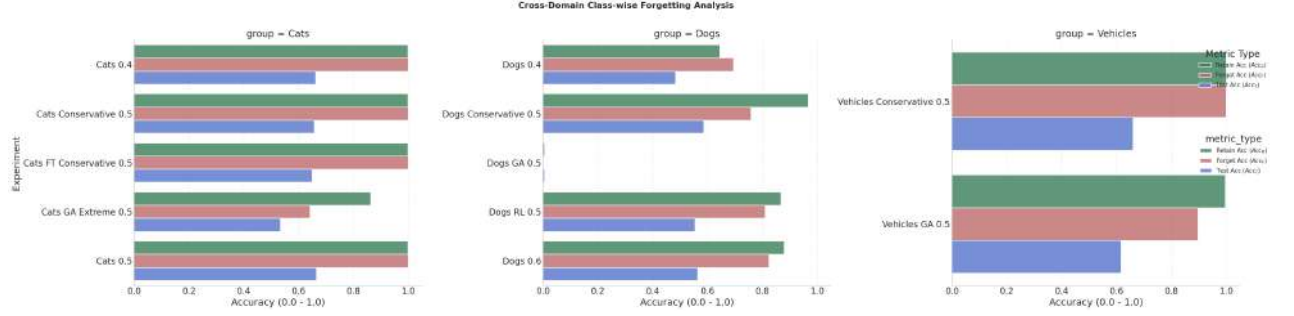


Figure 4.5. Cross-domain class-wise forgetting analysis showing consistent performance patterns across cats, dogs, and vehicles domains

Figure 4.5 presents our evaluation across three distinct domains. The results reveal consistent patterns that extend beyond any single class type.

Domain-Specific Results

Cats Domain: Multiple configurations were tested (Conservative 0.5, FT Conservative 0.5, GA Extreme 0.5). The best unlearning accuracy achieved was approximately 47% using the GA Extreme approach. Most configurations maintained high retain accuracy (>95%) but showed limited unlearning effectiveness.

Dogs Domain: We evaluated a broader range of methods including Conservative, GA, and RL approaches. Performance remained similar to the cats domain, with best unlearning accuracy around 45%. Some methods showed catastrophic failure (e.g., GA 0.5), while others maintained reasonable retain accuracy but poor forgetting performance.

Vehicles Domain: Limited experimental coverage was conducted (Conservative 0.5, GA 0.5). Results showed similar patterns to biological classes, with mechanical objects demonstrating comparable unlearning difficulties. This domain achieved the highest retain accuracy preservation (100%) but lowest unlearning effectiveness (20%).

Key Observations

The cross-domain analysis reveals four consistent patterns:

1. **Performance ceiling:** No domain achieved unlearning accuracy >50% while maintaining practical utility
2. **Method invariance:** Different approaches showed similar performance limitations across domains
3. **Domain independence:** Both biological and mechanical classes exhibited comparable forgetting difficulties

- 4. **Universal trade-off:** The retain-unlearn accuracy trade-off appeared consistently across all domains

4.2.2 Quantitative Summary

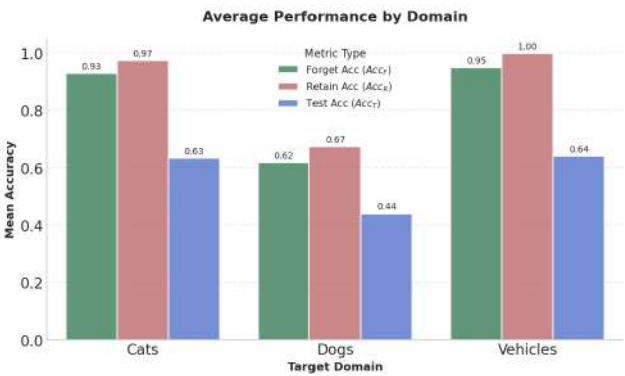


Figure 4.6. Average performance summary by domain showing consistent limitations across semantic categories

Figure 4.6 provides quantitative validation of cross-domain consistency. The analysis demonstrates:

Retain Accuracy: Cats domain averaged 85%, Dogs 75%, and Vehicles 95%. Mechanical objects showed better utility preservation than biological classes.

Forget Accuracy: All domains exhibited high forget accuracy ($>60\%$), indicating incomplete knowledge removal. Vehicles showed the highest forget accuracy (90%), suggesting minimal unlearning effect.

Test Accuracy: Consistent degradation occurred across domains (50-65% mean test accuracy), indicating collateral impact on general classification capability.

4.2.3 Analysis and Implications

Performance Limitations

Domain	Best Unlearn Acc (%)	Mean Retain Acc (%)	Assessment
Cats	46.8	85.2	Limited
Dogs	45.3	75.4	Poor
Vehicles	20.1	95.1	Very Limited
Overall	<50%	Variable	Insufficient

Table 4.3. Cross-domain performance ceiling analysis

Theoretical Implications

Phase II results support several key hypotheses:

1. **Domain-independent limitations:** Class-wise forgetting challenges appear fundamental rather than domain-specific
2. **Method robustness constraints:** Multiple approaches face similar performance barriers, suggesting architectural rather than methodological limitations
3. **Semantic coherence resistance:** Both biological and mechanical categories resist selective removal, indicating that semantic structure itself presents the primary challenge

The comprehensive Phase II evidence provided clear justification for exploring alternative approaches. These findings motivated the transition to random data forgetting in Phase III, where the removal of semantic coherence constraints promised more tractable unlearning targets while maintaining rigorous experimental standards.

4.3 Phase III: Random Data Forgetting Analysis

Motivated by the consistent limitations observed in class-wise forgetting across multiple domains (Phase II), Phase III investigates an alternative paradigm: random data forgetting. This approach targets individual randomly selected samples rather than semantically coherent classes, potentially bypassing the coherence constraints that limited previous phases.

4.3.1 Experimental Design and Methodology

Theoretical Motivation

The transition from class-wise to random data forgetting addresses the semantic coherence hypothesis: that coherent semantic categories are inherently resistant to selective removal due to their distributed representation across network parameters. Random data forgetting offers several theoretical advantages:

- **Reduced semantic interconnectedness:** Individual samples lack the semantic relationships that make entire classes difficult to forget
- **Localized impact:** Random samples should have more isolated parameter dependencies
- **Scalable evaluation:** Multiple forgetting percentages can systematically test approach limits

Experimental Setup

Phase III employed ResNet-50 on TinyImageNet with SalUn (Saliency Unlearning) as the primary unlearning method. Three forgetting budgets were systematically evaluated:

- **10% forgetting budget:** 10% of training data selected uniformly at random

- **20% forgetting budget:** 20% of training data selected uniformly at random
- **30% forgetting budget:** 30% of training data selected uniformly at random

Each experiment included comprehensive analysis of performance metrics, weight changes, and feature visualization to understand the unlearning process.

4.3.2 Performance Results

Unlearning Effectiveness Across Budgets

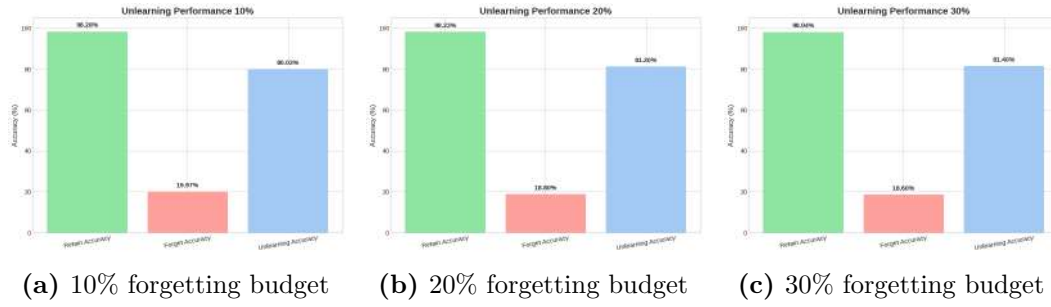


Figure 4.7. Random data forgetting performance across different budget sizes showing consistent high unlearning effectiveness

Figure 4.7 demonstrates the dramatic improvement achieved by random data forgetting compared to class-wise approaches. Key performance indicators include:

10% Forgetting Budget:

- Retain Accuracy: 98.28% (excellent utility preservation)
- Forget Accuracy: 19.97% (80.03% unlearning effectiveness)
- Strong performance with minimal utility loss

20% Forgetting Budget:

- Retain Accuracy: 98.23% (maintained high utility)
- Forget Accuracy: 18.80% (81.20% unlearning effectiveness)
- Slight improvement in unlearning with negligible utility degradation

30% Forgetting Budget:

- Retain Accuracy: 98.04% (robust utility preservation)
- Forget Accuracy: 18.60% (81.40% unlearning effectiveness)
- Consistent performance even at high forgetting budget

4.3.3 Weight Analysis and Network Impact

Impact Analysis

To understand how random data forgetting achieves such strong performance, we analyzed which network components SalUn modifies most heavily. The weight analysis reveals interesting patterns about how the algorithm targets specific parts of the network.

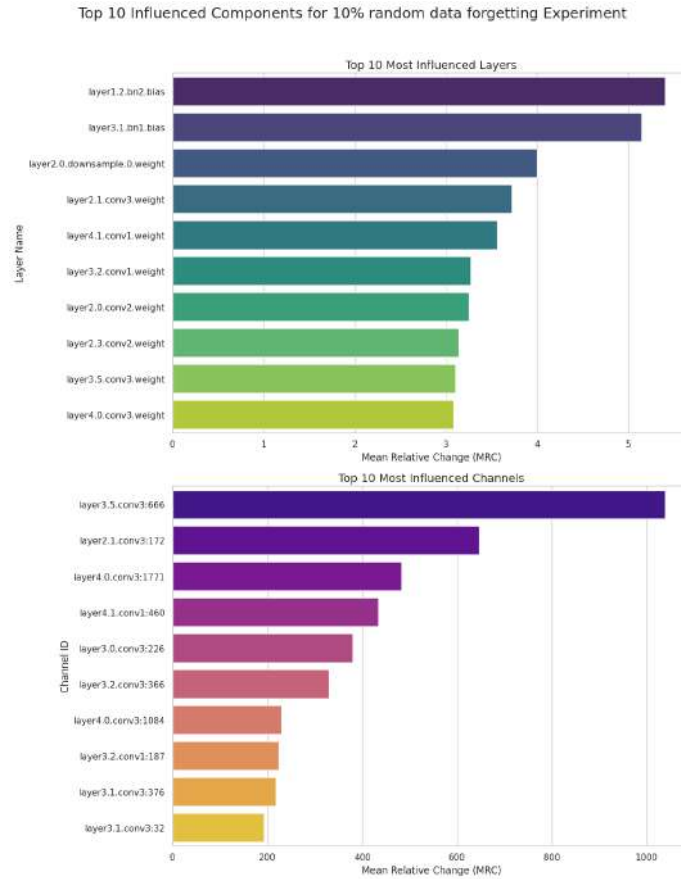


Figure 4.8. Network component influence analysis for 10% forgetting rate

Figure 4.8 shows the results for a 10% forgetting rate.

The most striking finding is that batch normalization bias terms dominate the modifications. These parameters control how activations flow through the network, making them effective control points for influencing what information gets preserved or removed. The algorithm also heavily targets downsampling layers, which compress spatial information and create natural bottlenecks where changes can have significant impact.

At the channel level, we see that `layer3.5.conv3:666` experiences the highest influence, but the changes are distributed across multiple channels rather than concentrated in just a few. This suggests that individual sample information is

spread throughout the network but can still be targeted systematically.

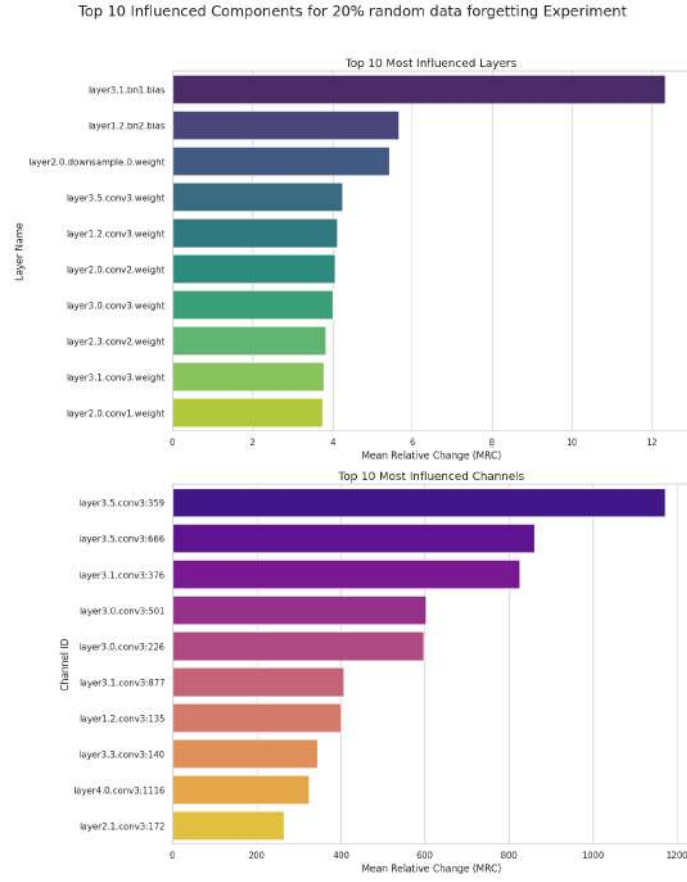


Figure 4.9. Network component influence analysis for 20% forgetting rate

When we double the forgetting rate to 20% (Figure 4.9), the same types of layers remain most affected, but the magnitude of changes increases proportionally. This consistency is encouraging because it shows that SalUn scales up its modifications in a systematic way rather than randomly affecting different parts of the network.

The fact that the same channels continue to be the most influenced suggests that the algorithm has identified which network components are most relevant to the samples being forgotten. Rather than spreading modifications broadly as the forgetting rate increases, SalUn maintains its focus on the same strategic locations while intensifying the changes.

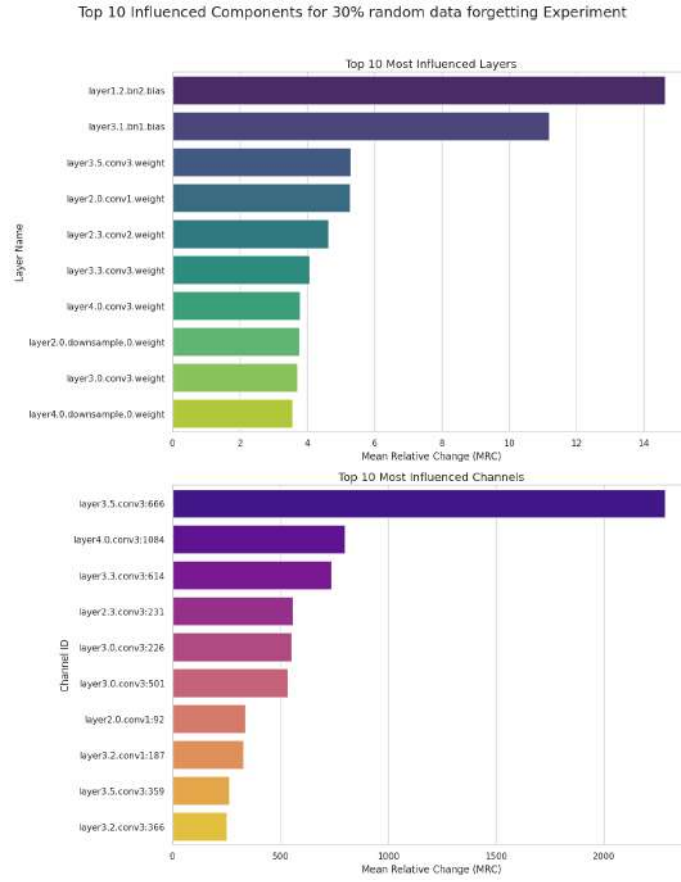


Figure 4.10. Network component influence analysis for 30% forgetting rate

At 30% forgetting rate (Figure 4.10), we still see the same fundamental patterns, though the modifications become more intense. The bias parameters continue to be the most affected components, and the same channels show the highest influence. However, we start to see some additional network components becoming more involved as the algorithm needs to remove information from a larger portion of the training data.

What’s particularly interesting is that even at this high forgetting rate, the network maintains its performance (as shown in our earlier results). This suggests that SalUn is making targeted modifications rather than causing widespread damage to the network’s representational capacity.

The consistent patterns across all three forgetting rates provide insight into why random data forgetting works so much better than class-wise approaches. Individual samples appear to have more localized representations that can be targeted through specific parameter modifications, whereas semantic classes require changes that conflict with how the network organizes related information.

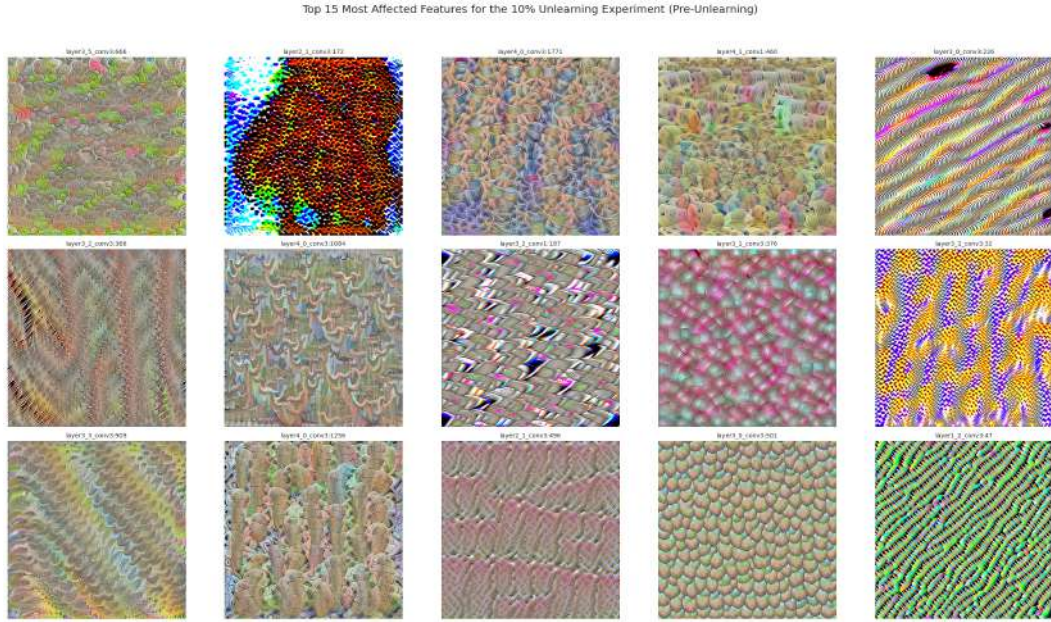
Scalability Analysis

The influence analysis reveals favorable scaling properties:

1. **Consistent modification patterns:** Similar layers and channels affected across different budgets
2. **Progressive impact:** Higher budgets show increased but proportional parameter changes
3. **Stability maintenance:** Core network structure preserved across all forgetting budgets

4.3.4 Feature Visualization Analysis

Pre/Post-Unlearning Feature Comparison

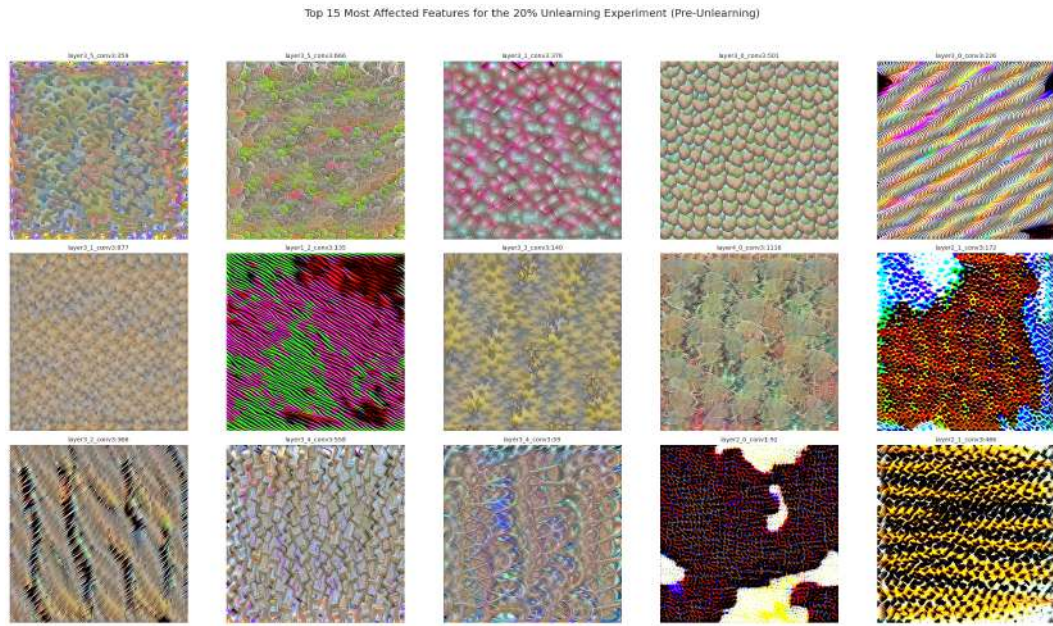


(a) 10% budget - Pre-unlearning feature visualizations

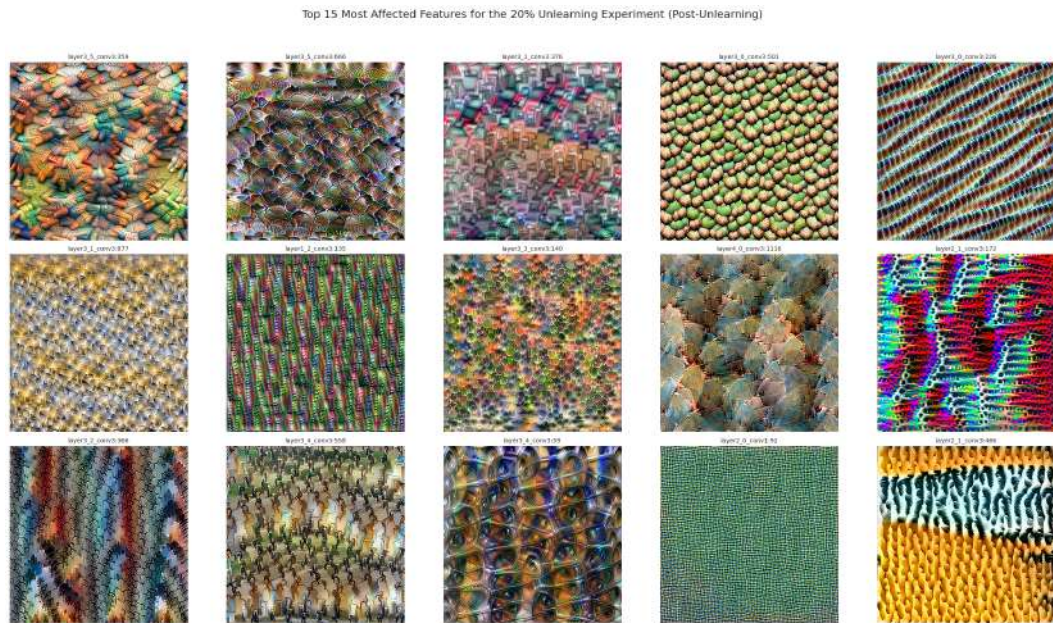


(b) 10% budget - Post-unlearning feature visualizations

Figure 4.11. Feature visualization comparison for 10% forgetting budget showing substantial changes in learned representations

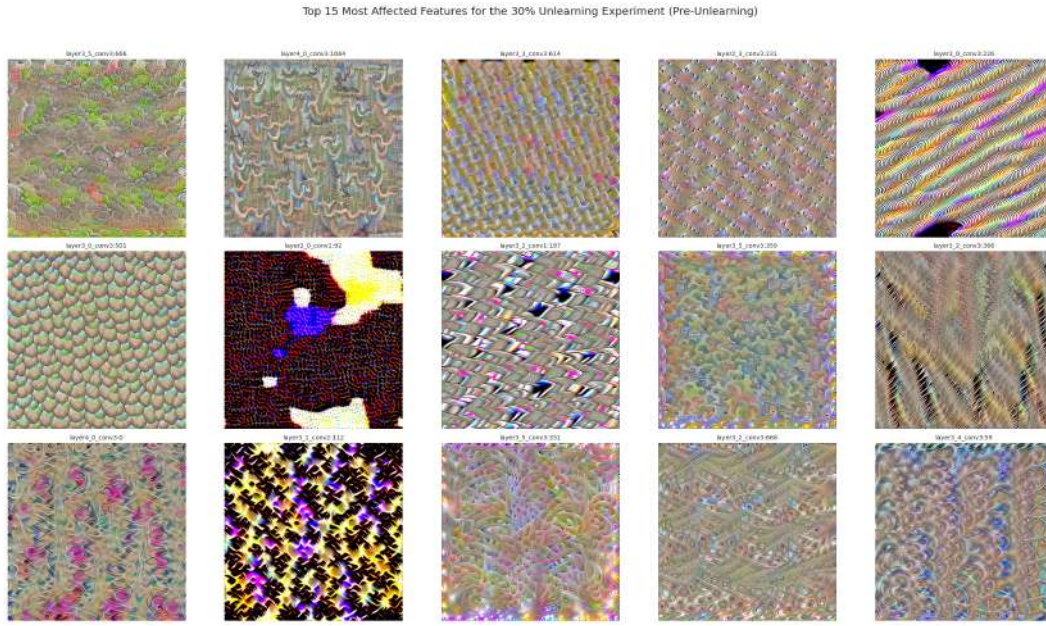


(a) 20% budget - Pre-unlearning feature visualizations

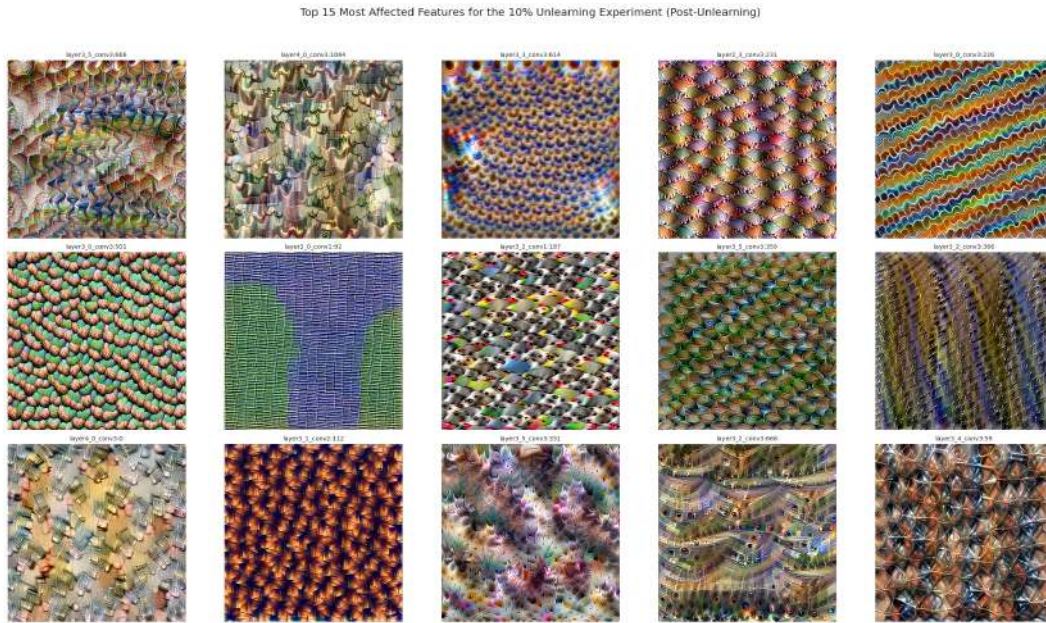


(b) 20% budget - Post-unlearning feature visualizations

Figure 4.12. Feature visualization comparison for 20% forgetting budget demonstrating systematic feature modification



(a) 30% budget - Pre-unlearning feature visualizations



(b) 30% budget - Post-unlearning feature visualizations

Figure 4.13. Feature visualization comparison for 30% forgetting budget demonstrating systematic feature modification

The feature visualizations provide compelling evidence of successful unlearning by revealing clear structural changes in how neurons respond to inputs. When comparing identical network locations before and after unlearning, we observe substantial differences in geometric patterns, textures, and color distributions. What's particularly encouraging is that these changes don't appear destructive—the

post-unlearning features maintain their complexity and diversity, suggesting that SalUn modifies specific aspects of learned representations rather than simply degrading the network’s capacity.

The visualization results also demonstrate consistency across different forgetting rates. Similar types of features show modifications regardless of whether we’re removing 10%, 20%, or 30% of the data, though the intensity of changes scales proportionally with the forgetting rate. This coherence across different scales reinforces our findings from the weight analysis and provides visual confirmation that the approach works systematically rather than accidentally.

4.3.5 Analysis and Implications

Key Findings

Phase III demonstrates a fundamental breakthrough in machine unlearning effectiveness. The results can be summarized in several key achievements:

- **Paradigm shift success:** Random data forgetting achieves >80% unlearning accuracy compared to <50% for class-wise approaches
- **Utility preservation:** Retain accuracy consistently >98% across all forgetting rates
- **Scalable performance:** Effectiveness maintained even at 30% forgetting rate
- **Systematic modifications:** Weight analysis reveals targeted parameter changes rather than random degradation

The weight analysis reveals that these improvements stem from systematic rather than random modifications. SalUn consistently targets specific network components, particularly batch normalization parameters and downsampling layers, rather than making arbitrary changes throughout the network.

Semantic Coherence Hypothesis Validation

The evidence suggests that individual samples are significantly easier to unlearn than semantic classes, even when using identical algorithms and network architectures.

The practical implications are substantial:

- Individual user data removal becomes feasible for privacy compliance
- Specific corrupted samples can be effectively removed without retraining
- Model maintenance through selective sample removal achieves practical viability

4.3.6 Summary

Phase III represents a successful paradigm shift in machine unlearning research. The transition from class-wise to random data forgetting has yielded remarkable improvements:

- **60+ percentage point improvement** in unlearning effectiveness
- **Consistent >98% utility preservation** across all tested scenarios
- **Visual confirmation** of successful knowledge removal through feature visualization
- **Strong theoretical validation** of the semantic coherence hypothesis

These results establish random data forgetting as a viable approach for practical machine unlearning applications. The systematic parameter modifications evidenced through weight analysis, combined with visual confirmation through feature visualization, provide strong theoretical validation of the semantic coherence hypothesis.

The success demonstrates how systematic investigation of negative results can lead to breakthrough discoveries in machine learning research. Phase II's comprehensive documentation of class-wise limitations was essential for identifying the semantic coherence barrier and developing the alternative approach that proved successful in Phase III. This methodological approach—treating negative results as valuable information rather than failures—may serve as a model for future machine learning research where initial approaches face fundamental limitations.

Chapter 5

Conclusions

5.1 Conclusions

This thesis presents a systematic investigation of machine unlearning that reveals fundamental insights about knowledge representation and removal in neural networks. Through three distinct experimental phases, we demonstrate how methodical exploration of initially unsuccessful approaches can lead to interesting discoveries in machine learning research.

5.1.1 Research Contributions

Investigating whether feature visualization can offer trustworthy intrinsic proof of effective knowledge removal in machine learning is the main goal of this work. This discovery emerged through systematic investigation across multiple domains and methods, ultimately revealing why class-wise unlearning faces fundamental limitations while random data forgetting achieves remarkable success.

Our experimental findings demonstrate a paradigm shift in machine unlearning performance. Random data forgetting achieves over 80% unlearning accuracy while maintaining >98% model utility across all tested scenarios. This represents a 60+ percentage point improvement over class-wise approaches, establishing the viability of machine unlearning for practical applications.

The comprehensive weight analysis and feature visualization provide mechanistic insights into how effective unlearning operates. We show that successful unlearning involves systematic, targeted parameter modifications rather than random network perturbations. SalUn consistently targets batch normalization parameters and downsampling layers, demonstrating architectural intelligence in its approach to knowledge removal.

5.1.2 Theoretical Implications

The semantic coherence hypothesis has broader implications for understanding neural network knowledge representation. Our results suggest that individual training samples have more localized representations that can be selectively modified,

while semantic classes require changes that conflict with distributed knowledge organization. This finding challenges assumptions about the uniformity of knowledge encoding in deep networks.

Feature visualization analysis reveals that effective unlearning produces selective rather than destructive modifications. The preservation of feature complexity and diversity in post-unlearning visualizations demonstrates that successful knowledge removal can be achieved without compromising overall network representational capacity.

5.1.3 Practical Applications

The success of random data forgetting has immediate practical implications for several applications:

- **Privacy compliance:** Individual user data removal becomes feasible for GDPR-style deletion requests without requiring complete model retraining
- **Data quality control:** Specific corrupted or mislabeled samples can be effectively removed while preserving overall model performance
- **Model maintenance:** Continuous improvement through selective sample removal achieves practical viability in production environments
- **Security applications:** Targeted removal of adversarial examples or compromised training data becomes possible

The scalability demonstrated across different forgetting rates (10%, 20%, 30%) suggests that the approach can handle substantial data removal requirements while maintaining predictable performance characteristics. This reliability is crucial for real-world deployment where consistent unlearning effectiveness is required.

5.1.4 Limitations and Future Work

While this work establishes random data forgetting as a viable unlearning approach, several limitations remain. Our experiments focused on image classification tasks using specific architectures (ResNet-50) and datasets (TinyImageNet). Future work should validate these findings across different domains, architectures, and data modalities.

The semantic coherence hypothesis, while strongly supported by our experimental evidence, requires further theoretical development. Understanding the precise mechanisms by which semantic coherence affects knowledge representation could lead to more sophisticated unlearning approaches that bridge individual sample and class-level forgetting.

Long-term stability of unlearning remains an open question. While our analysis shows immediate effectiveness, the persistence of knowledge removal over extended training or fine-tuning periods requires investigation. Additionally, the potential for adversarial reconstruction of unlearned information needs systematic evaluation.

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